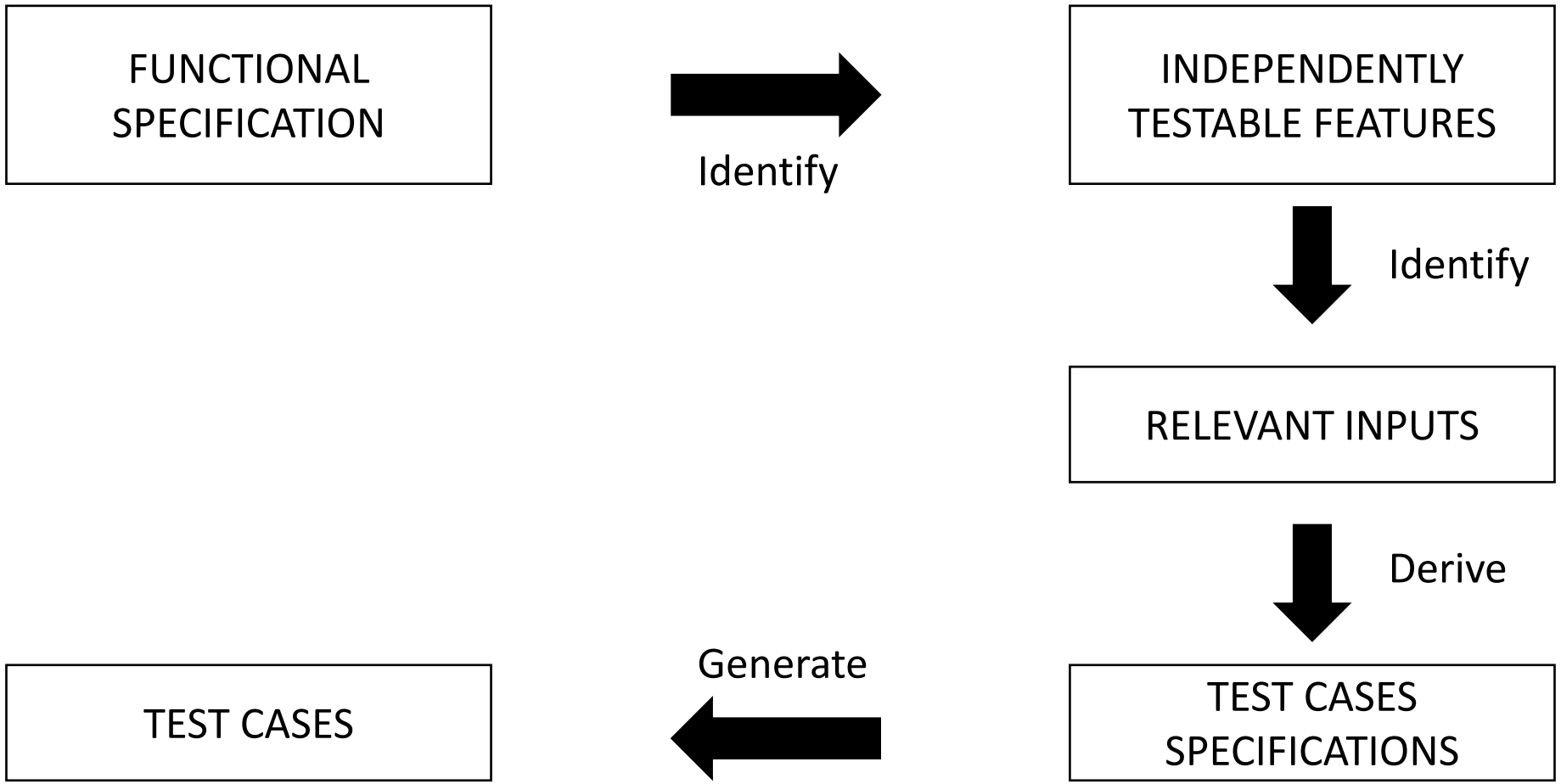
CS3300 A: Introduction to Software Engineering

Lecture 15: Black-Box Testing; TOTT#7: AI-Assisted Black-Box API Testing.

Dr. Nimisha Roy ▸ nroy9@gatech.edu

Slides adapted from Alessandro Orso

A systematic Functional-Testing Approach

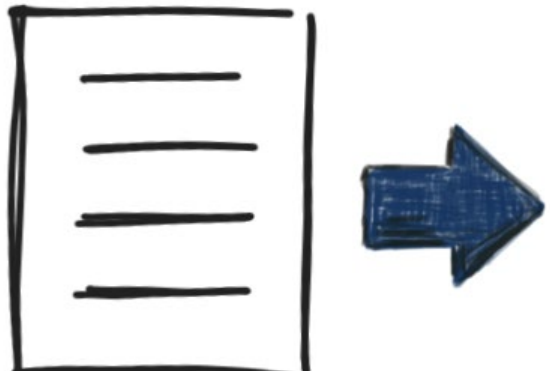


Decoupling; Automated Sub-tasks; Monitor testing process

Slide adapted from Alessandro Orso

A systematic Functional-Testing Approach to Black-box Testing

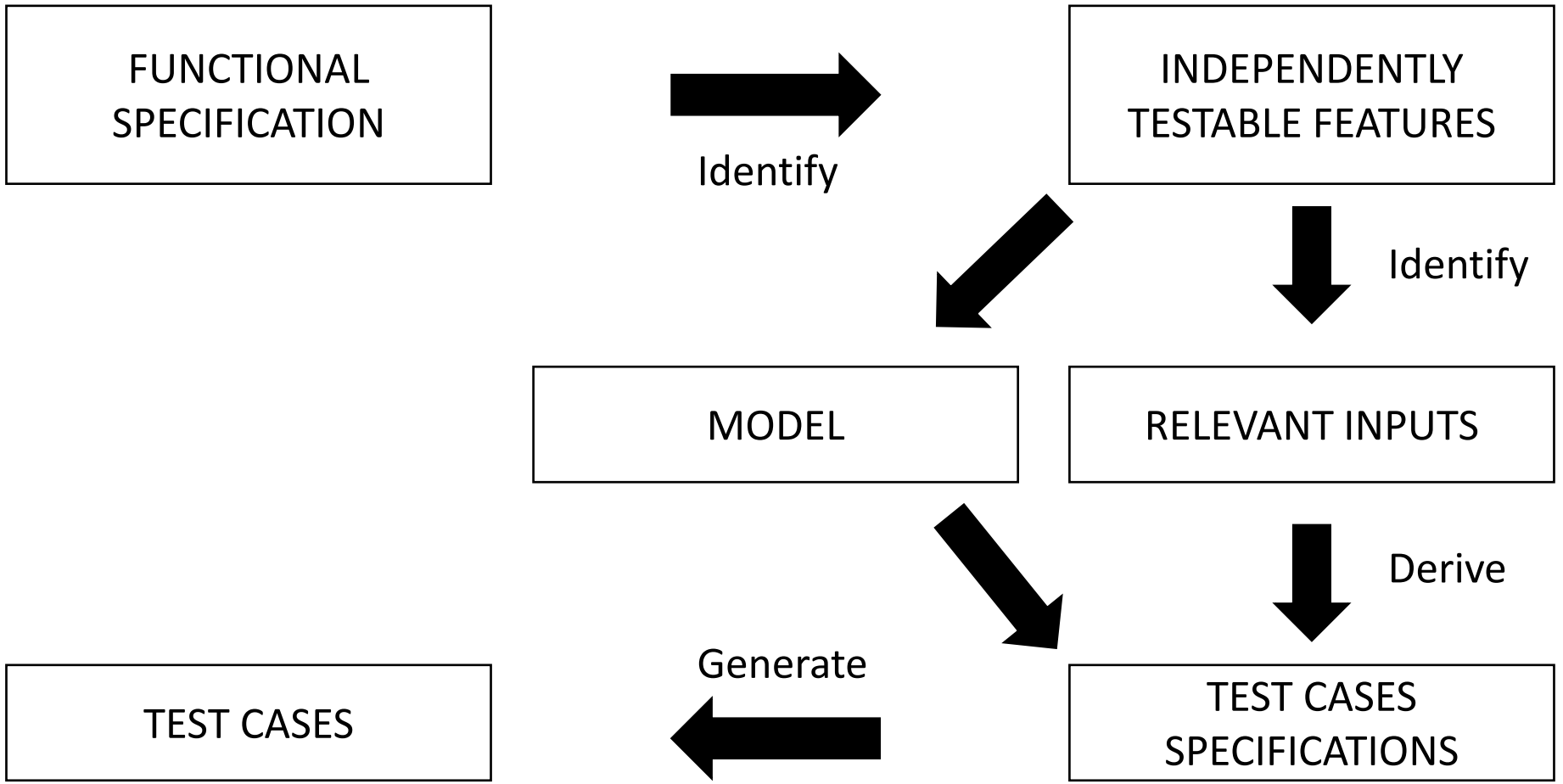
The Category-Partition Method



Functional Specification

1. Identify independently testable features
2. Identify Categories (characteristic features of input)
3. Partition Categories into choices (interesting values – boundary values)
4. Identify constraints among choices (PROPERTY---- IF, ERROR, SINGLE)
5. Produce/Evaluate test case specifications (Produce test frames, can be automated using TSLGenerator)
6. Generate test cases from test case specifications (by instantiating)

Model-Based Testing



Slide adapted from Alessandro Orso

Model-Based Testing



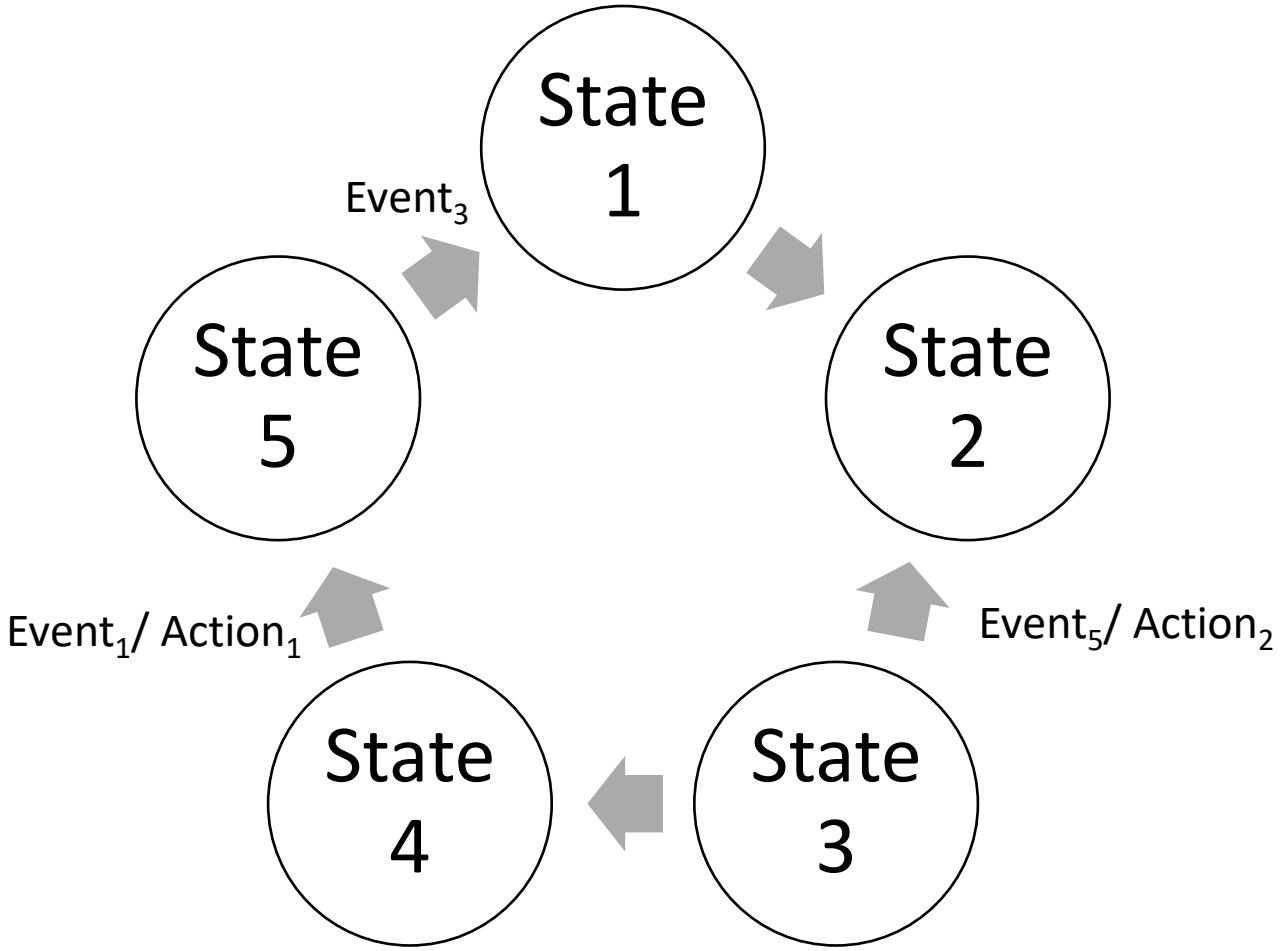
Specification

Model
(Abstract representation
of the software)

Test Cases

Finite State Machines (FSM)

Nodes = States
Edges = Transitions
Edge Labels = Events/Actions

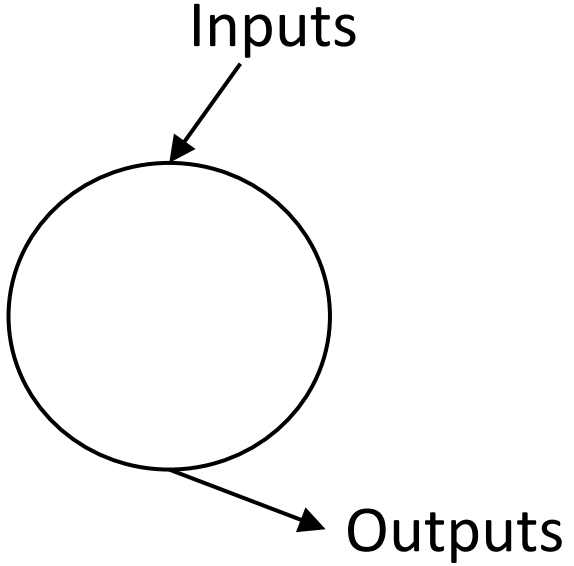


Building an FSM

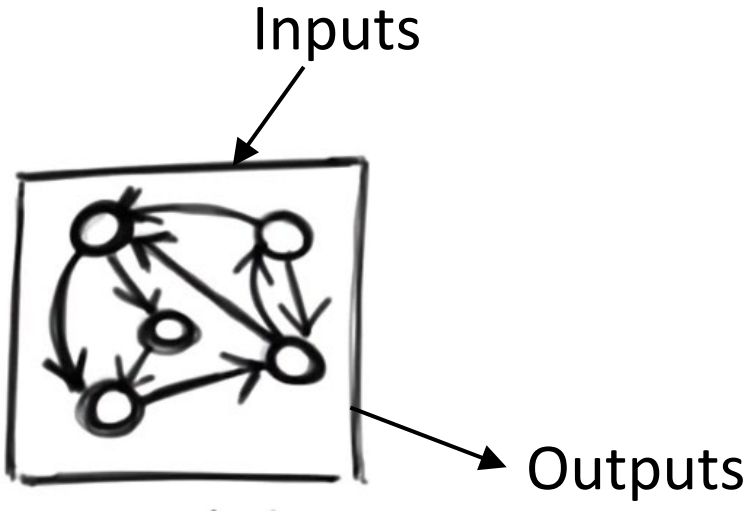


Specification

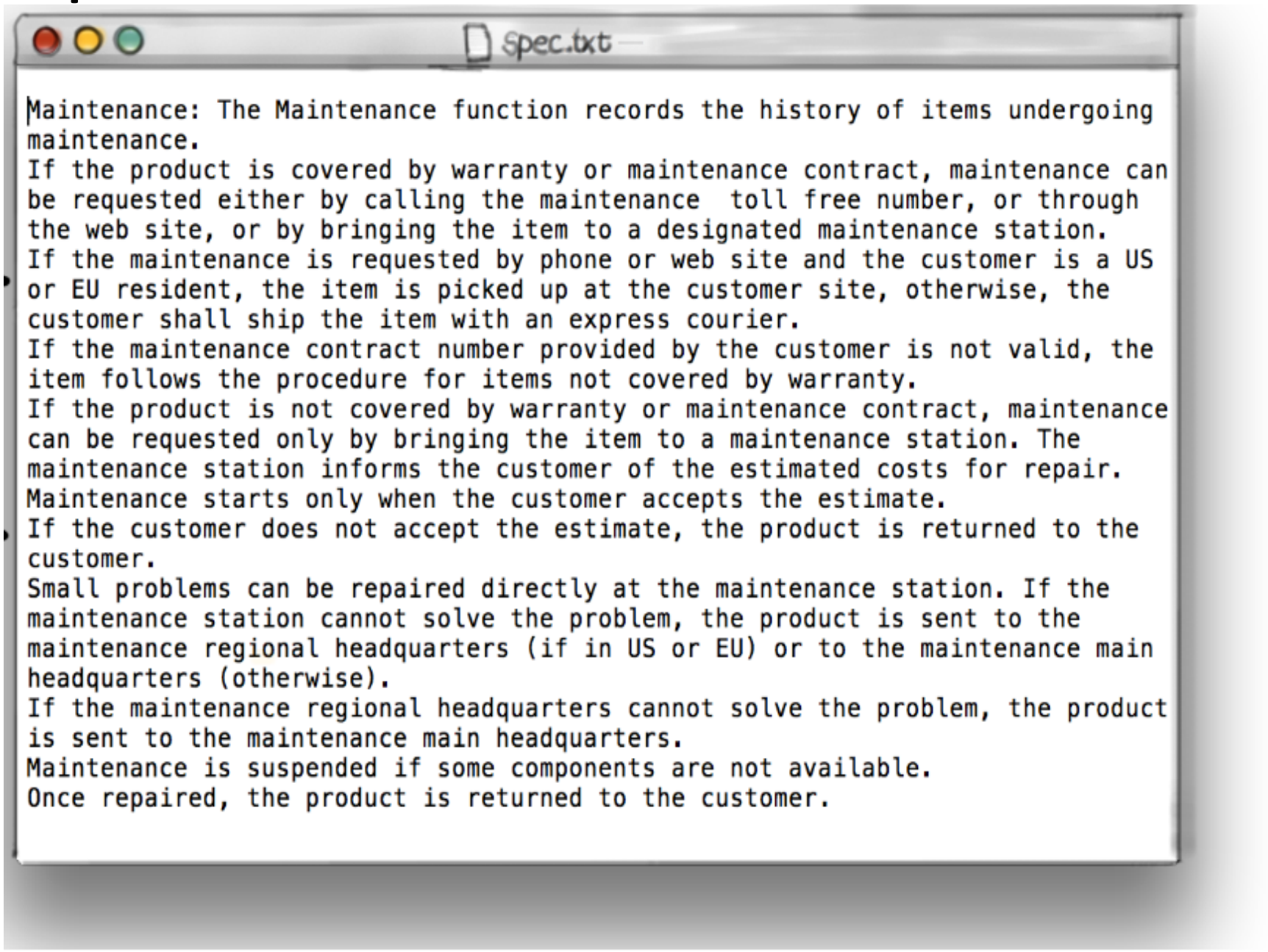
Identify System's Boundaries, and Input and Output



Identify relevant states and transitions

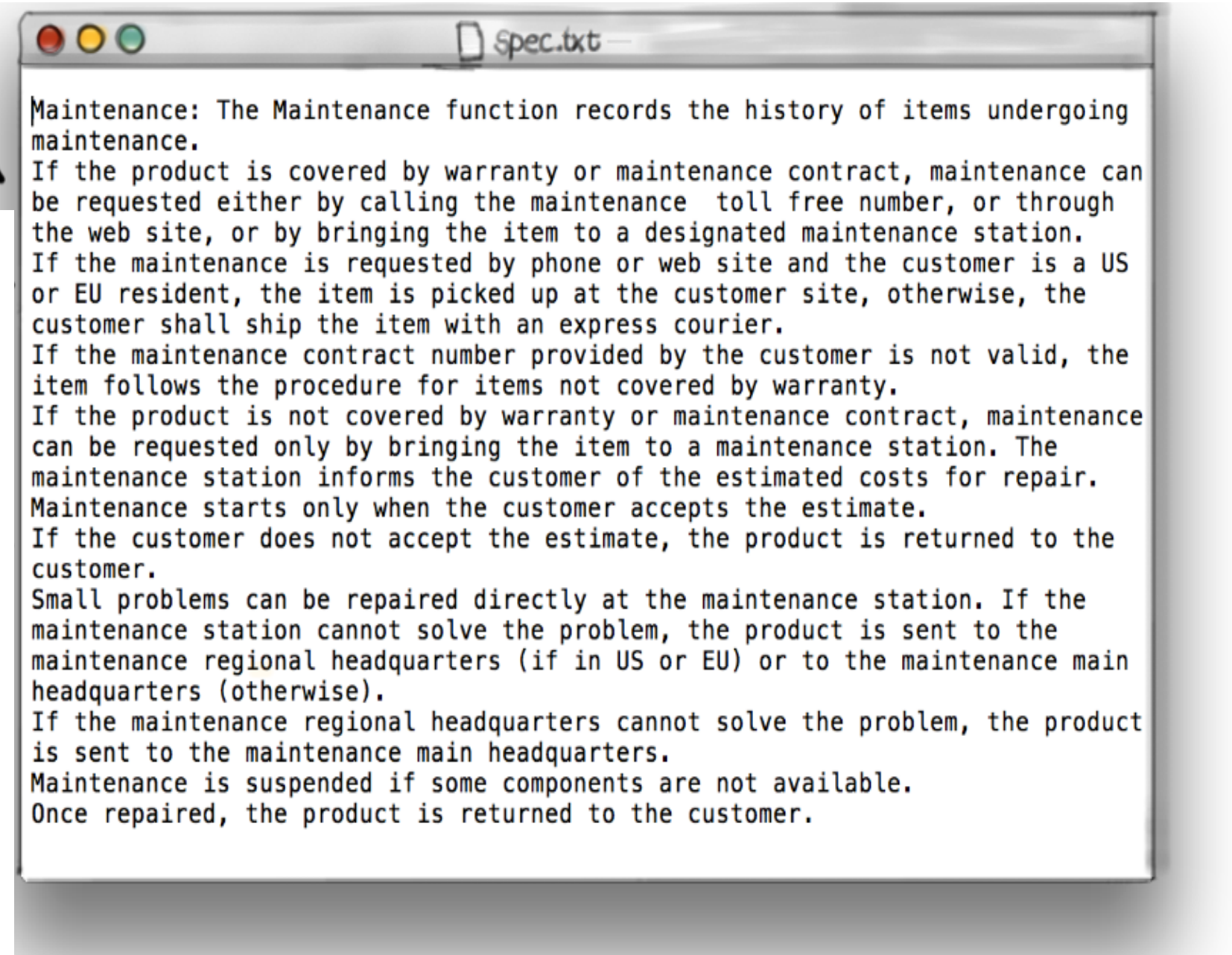


From an Informal Specification…

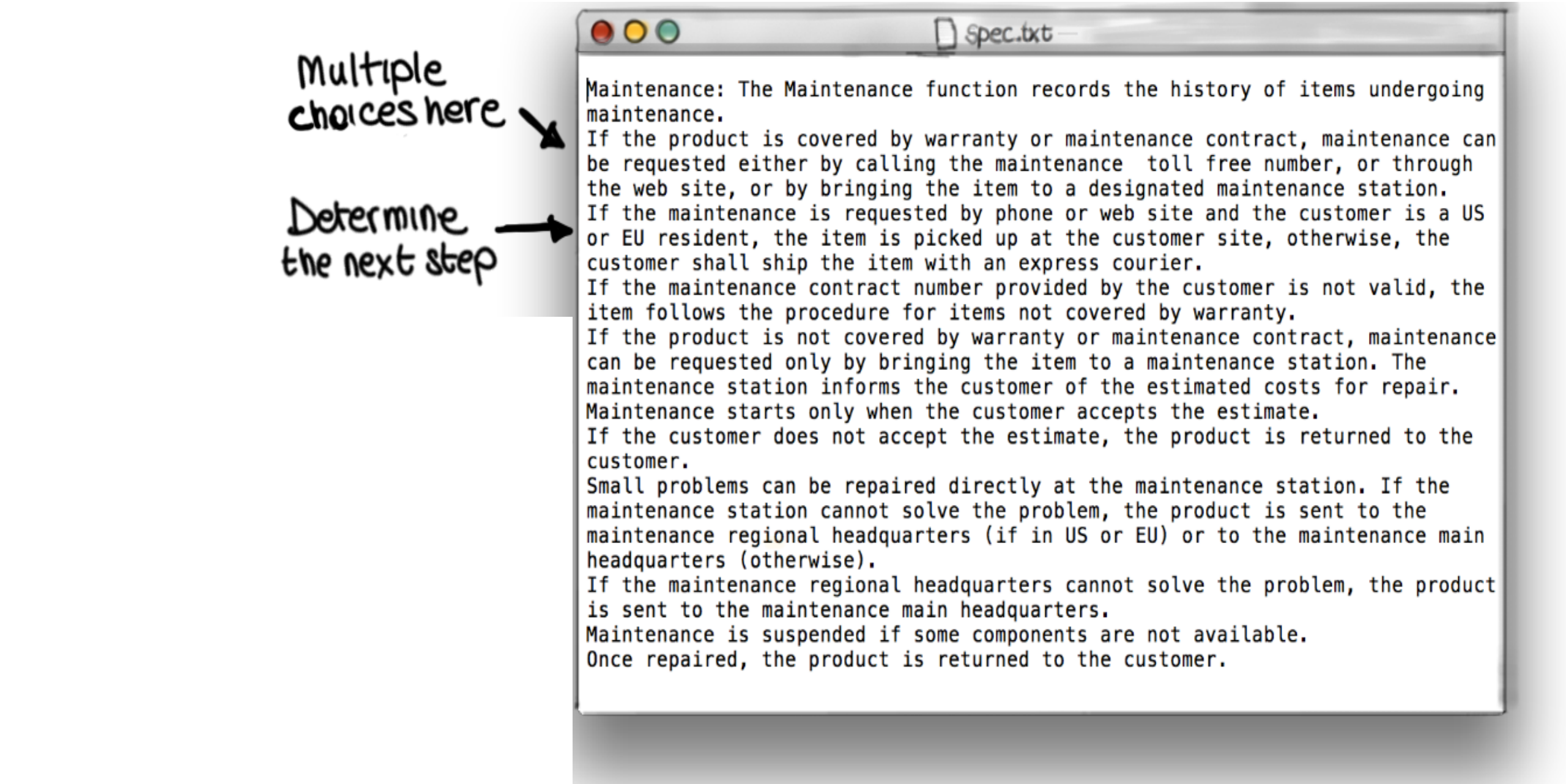


From an Informal Specification…

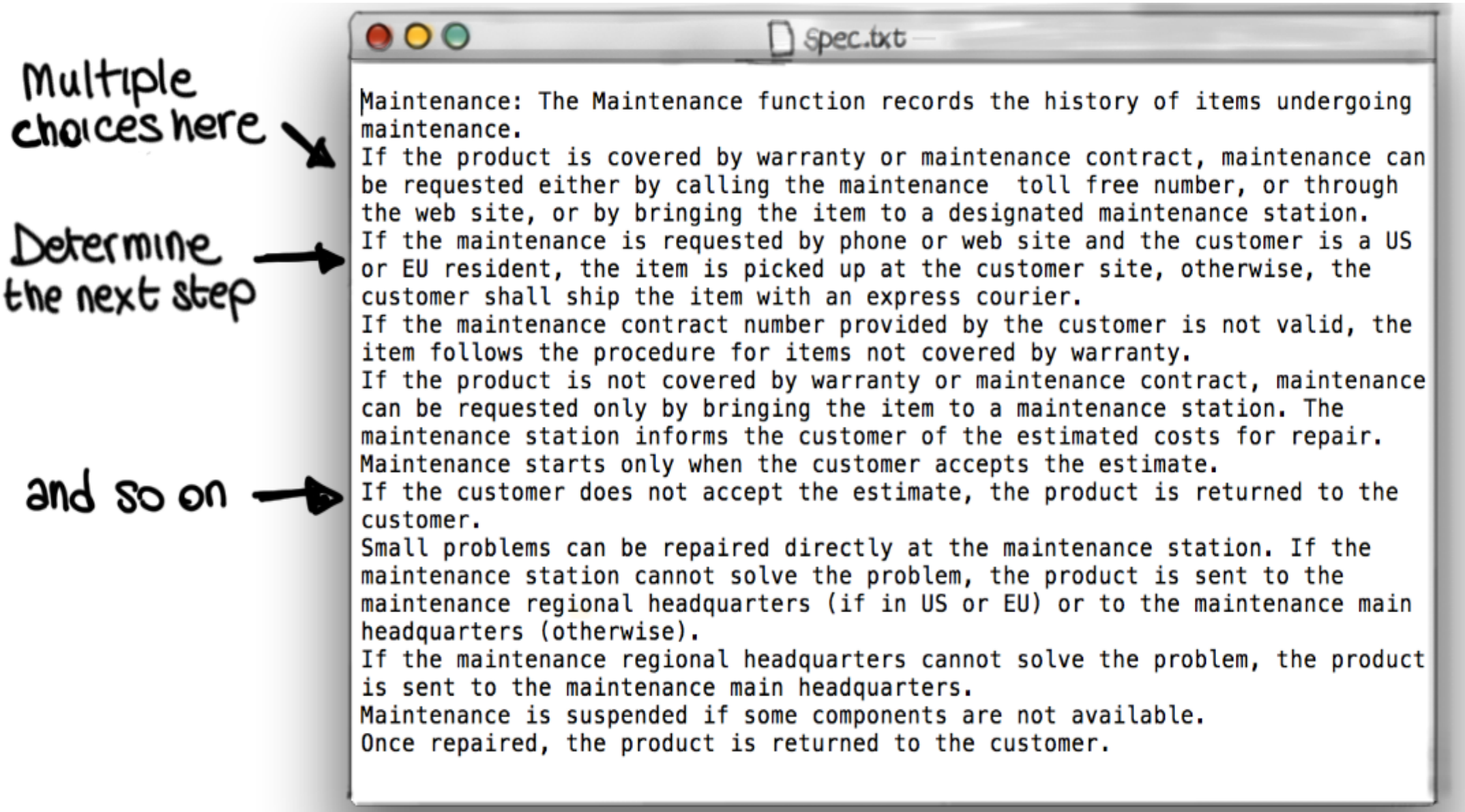
Multiple choices here



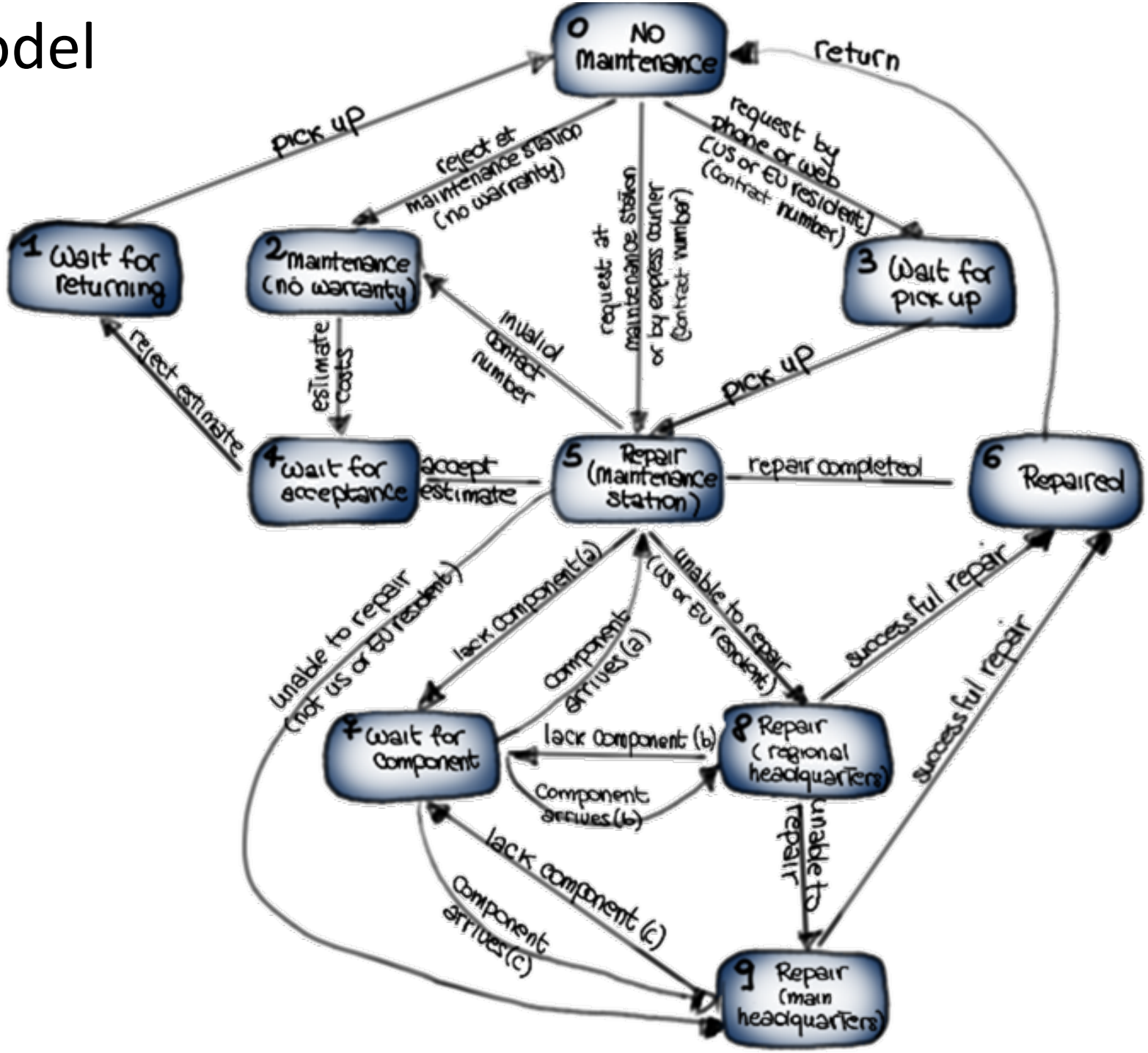
From an Informal Specification…



From an Informal Specification…



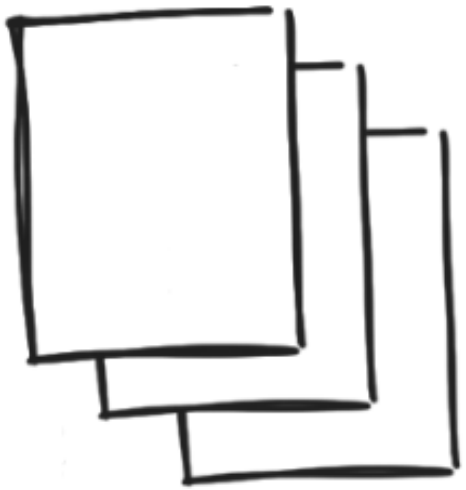
… To a Finite State Model



Finite State Model to a Set of Test Cases





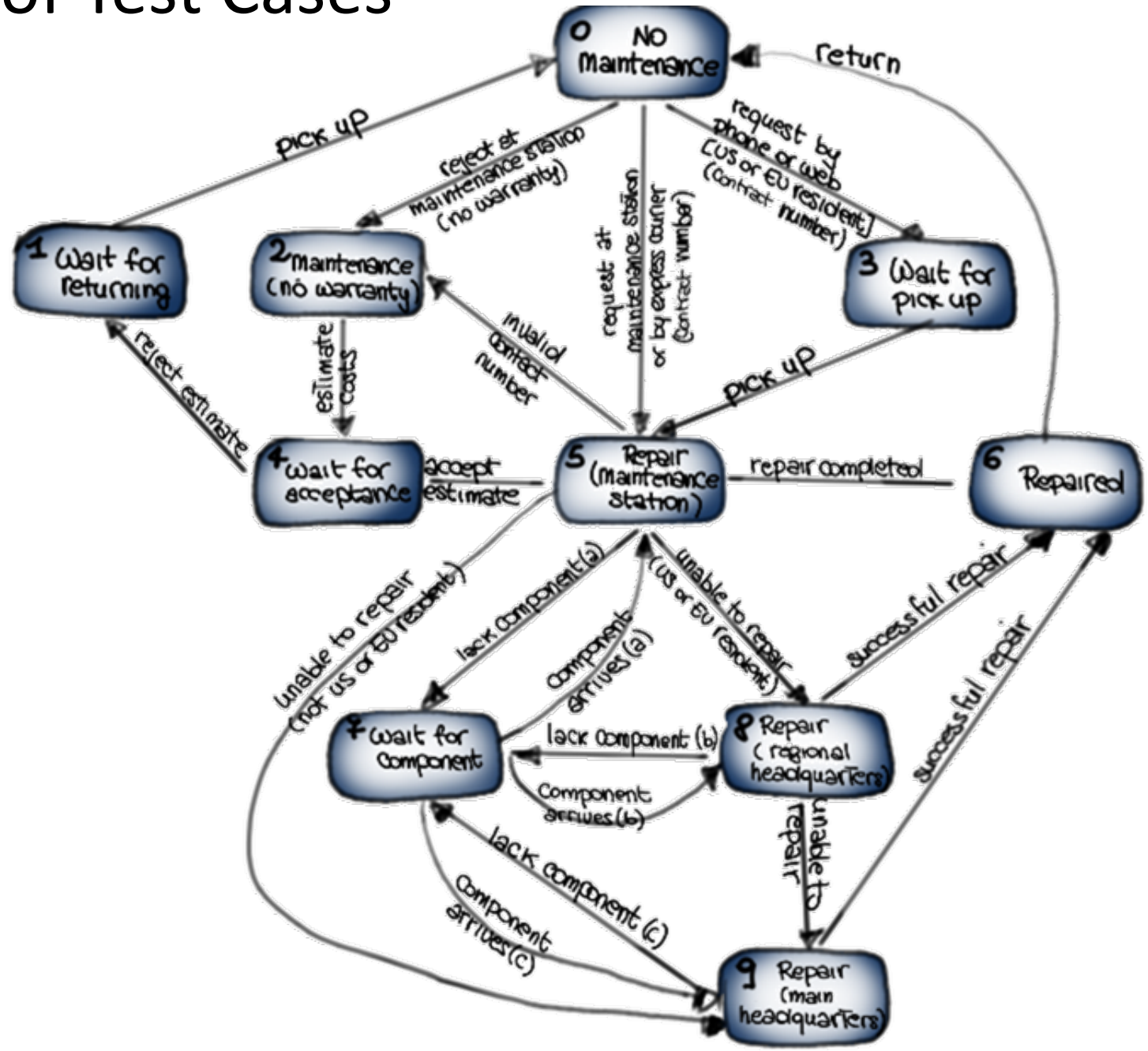


Test Cases

Finite State Model to a Set of Test Cases

Cover the behaviors represented by the state machine.

- Cover all the states
- Or Identify paths in state machine that go through all states in the machine

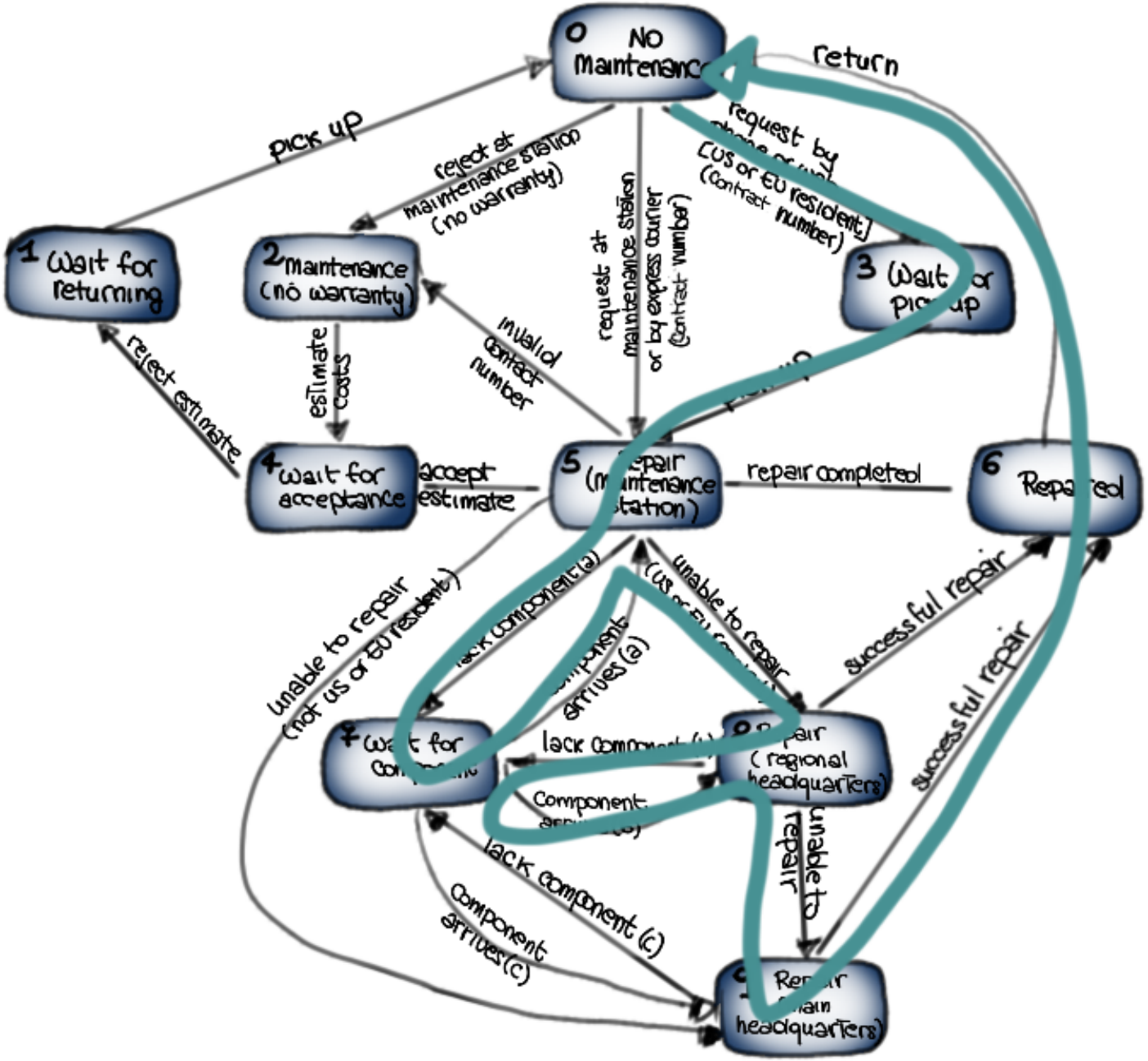


Finite State Model to a Set of Test Cases

Cover the behaviors represented by the state machine.

- Cover all the states
- Or Identify paths in state machine that go through all states in the machine

TC1: Φ, 3, 5, 7, 5, 8, 7, 8, 9, 6, Φ



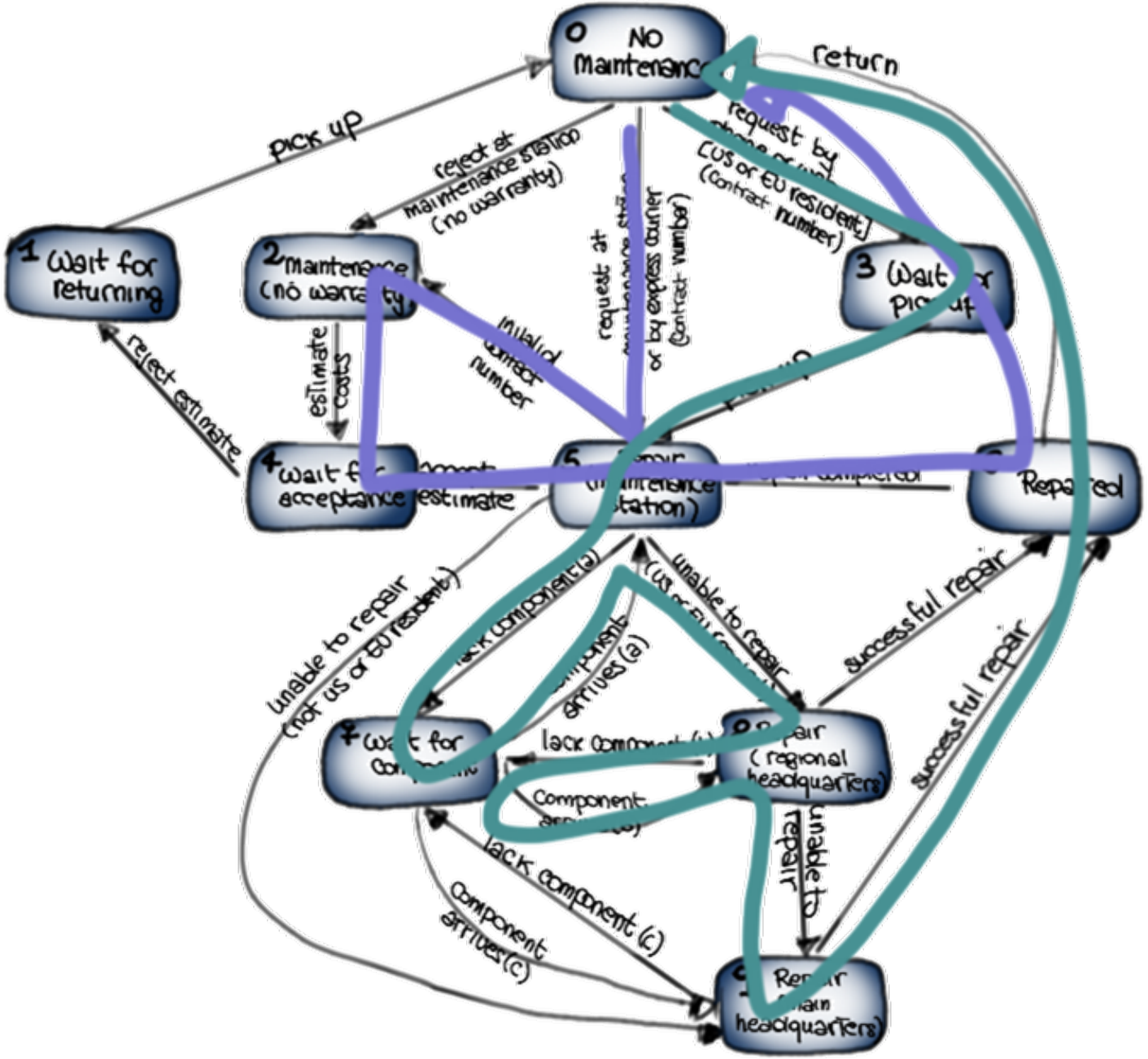
Finite State Model to a Set of Test Cases

Cover the behaviors represented by the state machine.

- Cover all the states
- Or Identify paths in state machine that go through all states in the machine

TC1: Φ, 3, 5, 7, 5, 8, 7, 8, 9, 6, Φ

TC2: Φ, 5, 2, 4, 5, 6, Φ



Finite State Model to a Set of Test Cases

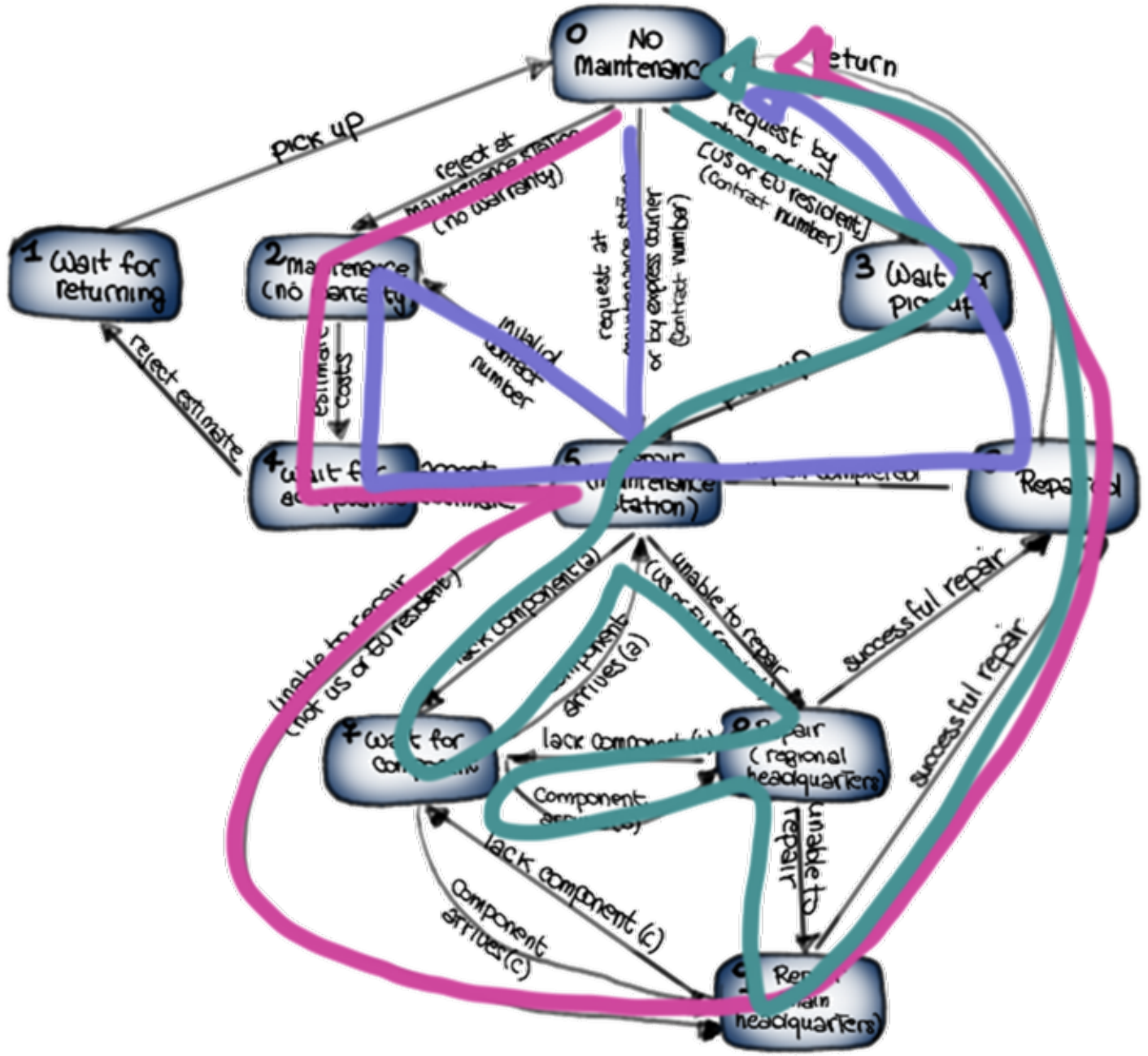
Cover the behaviors represented by the state machine.

- Cover all the states
- Or Identify paths in state machine that go through all states in the machine

TC1: Φ, 3, 5, 7, 5, 8, 7, 8, 9, 6, Φ

TC2: Φ, 5, 2, 4, 5, 6, Φ

TC3: Φ, 2, 4, 5, 9, 6, Φ



Finite State Model to a Set of Test Cases

Cover the behaviors represented by the state machine.

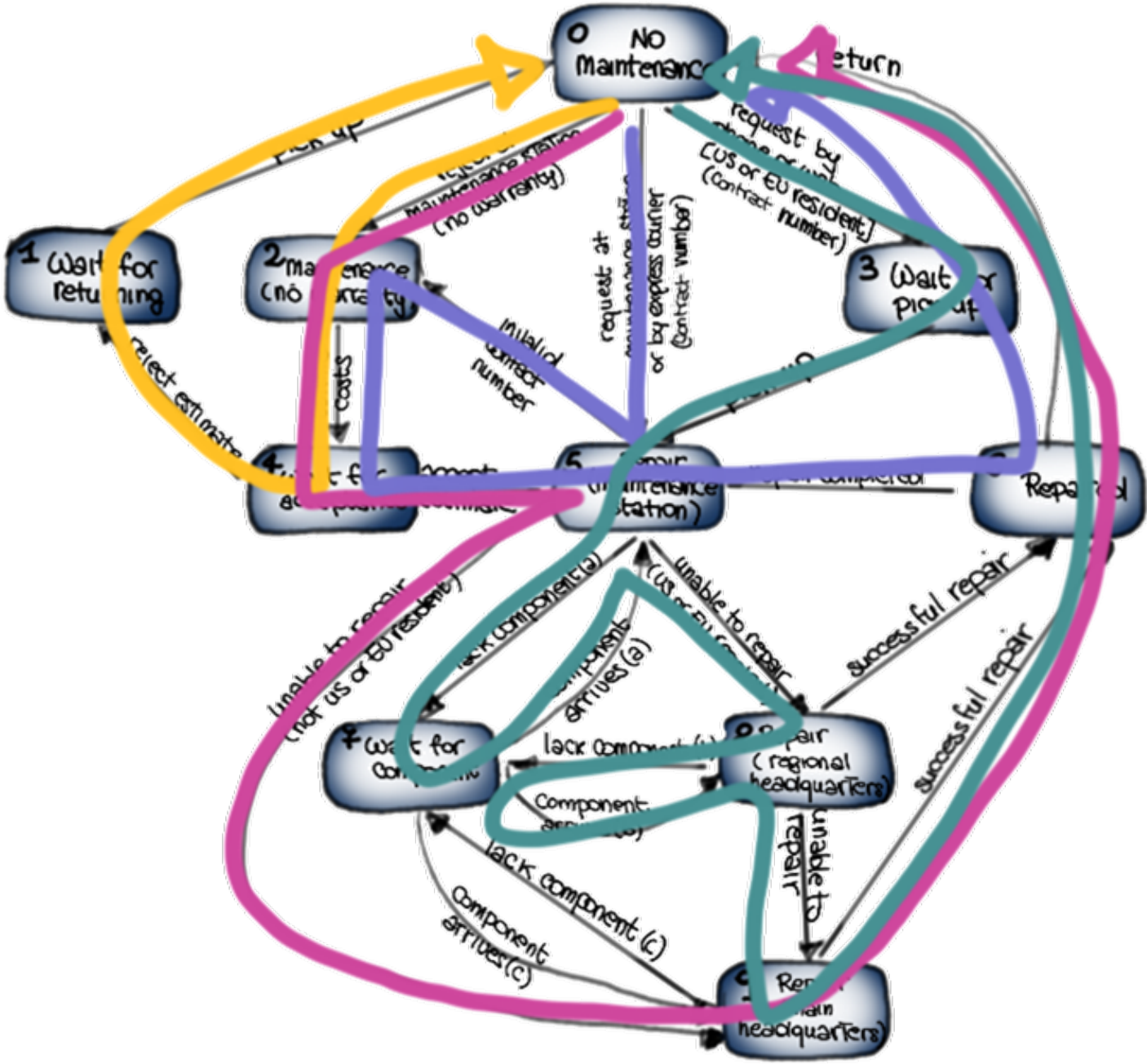
- Cover all the states
- Or Identify paths in state machine that go through all states in the machine
- Or cover all transitions

TC1: Φ, 3, 5, 7, 5, 8, 7, 8, 9, 6, Φ

TC2: Φ, 5, 2, 4, 5, 6, Φ

TC3: Φ, 2, 4, 5, 9, 6, Φ

TC4: Φ, 2, 4, 1, Φ



Finite State Model to a Set of Test Cases

Cover the behaviors represented by the state machine.

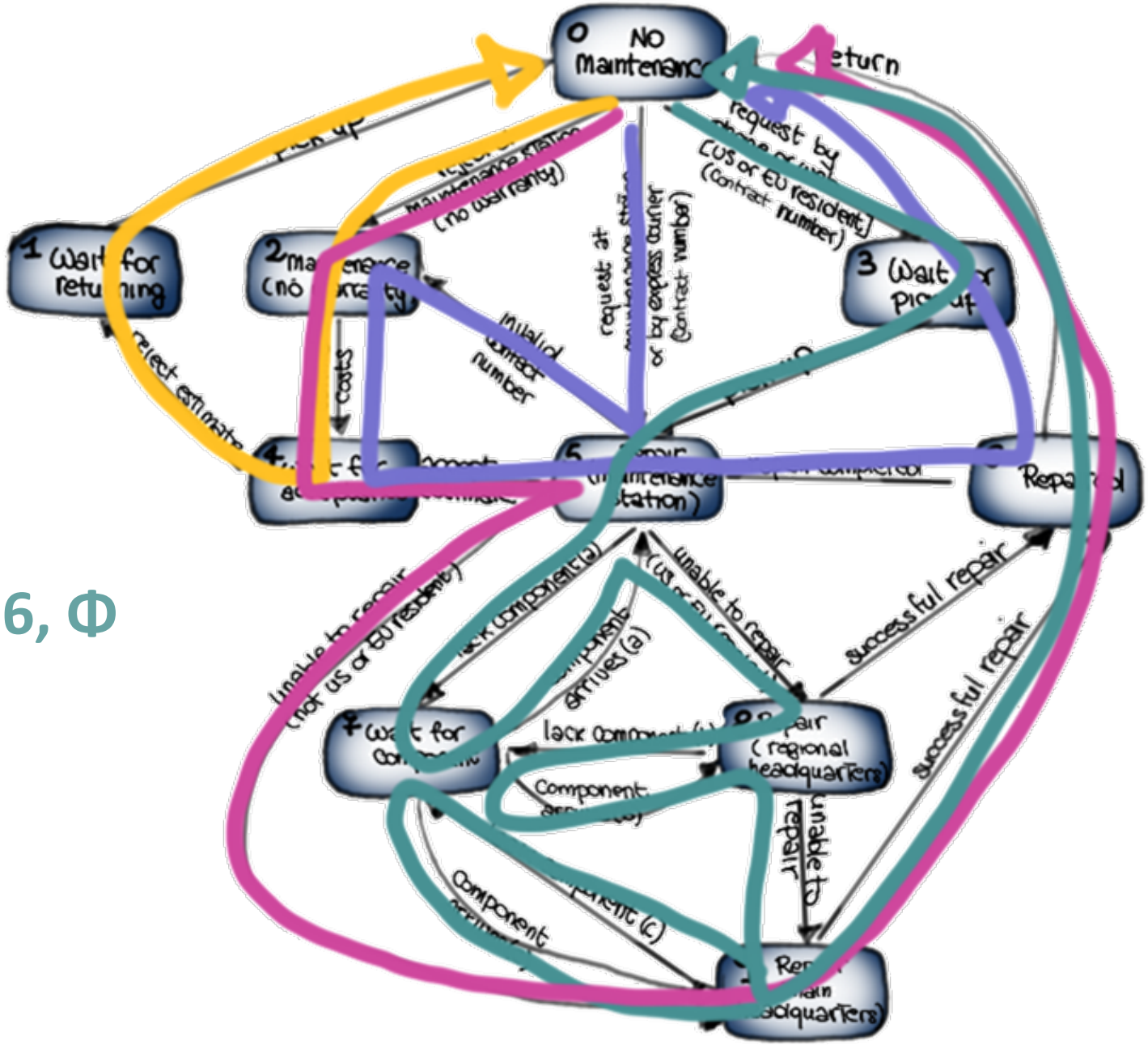
- Cover all the states
- Or Identify paths in state machine that go through all states in the machine
- Or cover all transitions

TC1: Φ, 3, 5, 7, 5, 8, 7, 8, 9, 7, 9, 6, Φ

TC2: Φ, 5, 2, 4, 5, 6, Φ

TC3: Φ, 2, 4, 5, 9, 6, Φ

TC4: Φ, 2, 4, 1, Φ



Finite State Model to a Set of Test Cases

Cover the behaviors represented by the state machine.

- Cover all the states
- Or Identify paths in state machine that go through all states in the machine
- Or cover all transitions

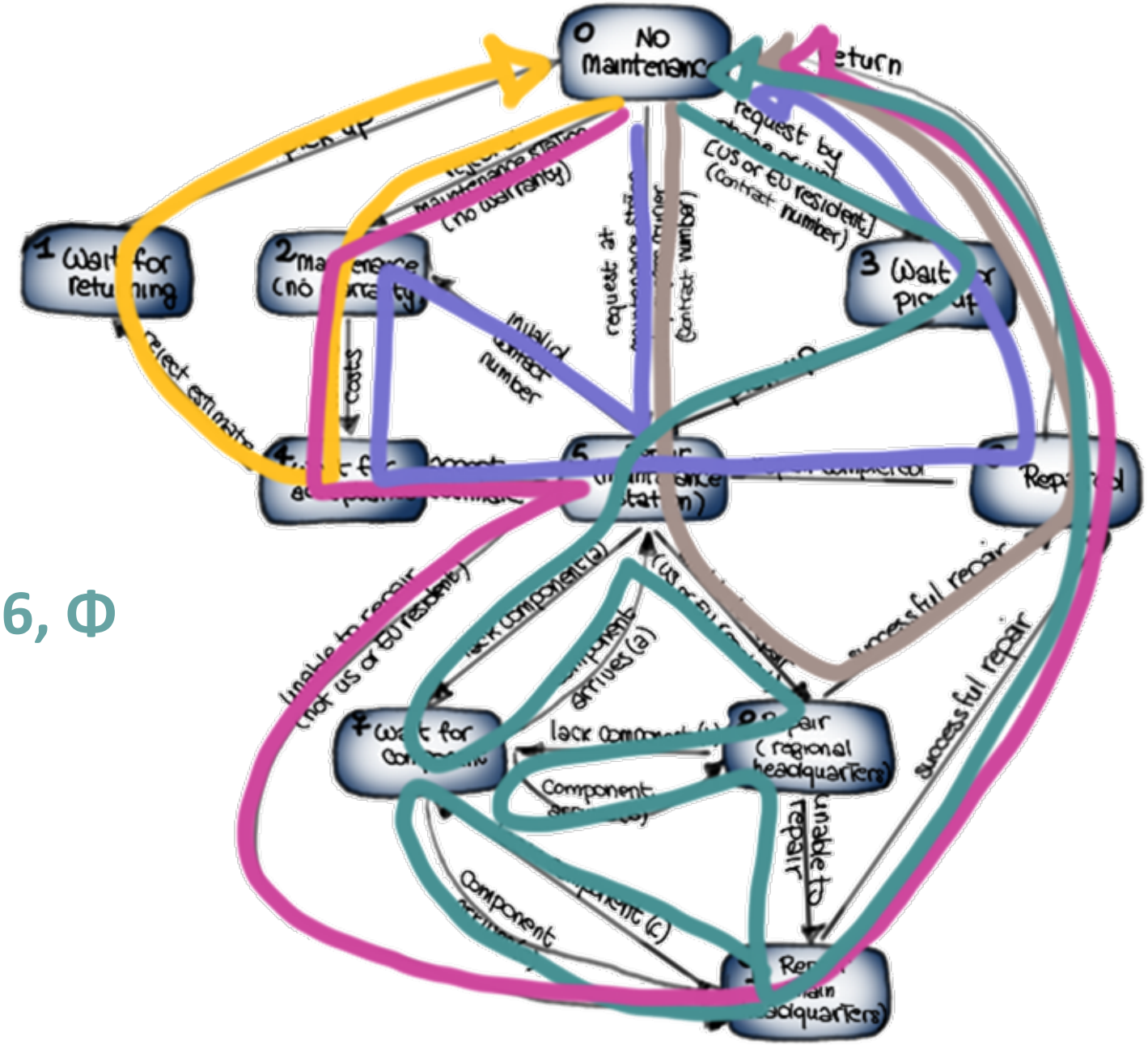
TC1: Φ, 3, 5, 7, 5, 8, 7, 8, 9, 7, 9, 6, Φ

TC2: Φ, 5, 2, 4, 5, 6, Φ

TC3: Φ, 2, 4, 5, 9, 6, Φ

TC4: Φ, 2, 4, 1, Φ

TC5: Φ, 5, 8, 6, Φ



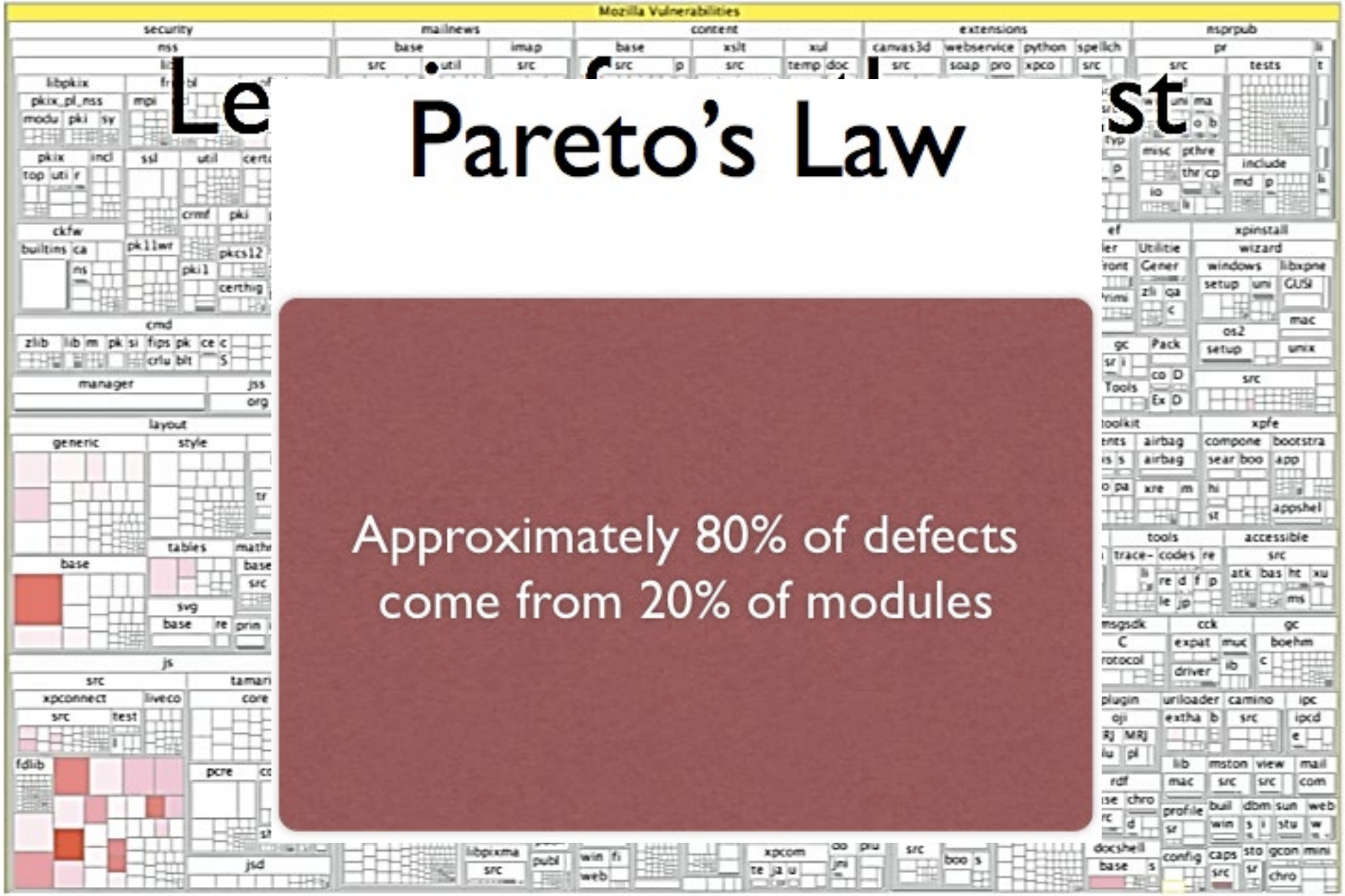
Some Considerations

Applicability
- Very General Approach
- In UML, state machine are readily available

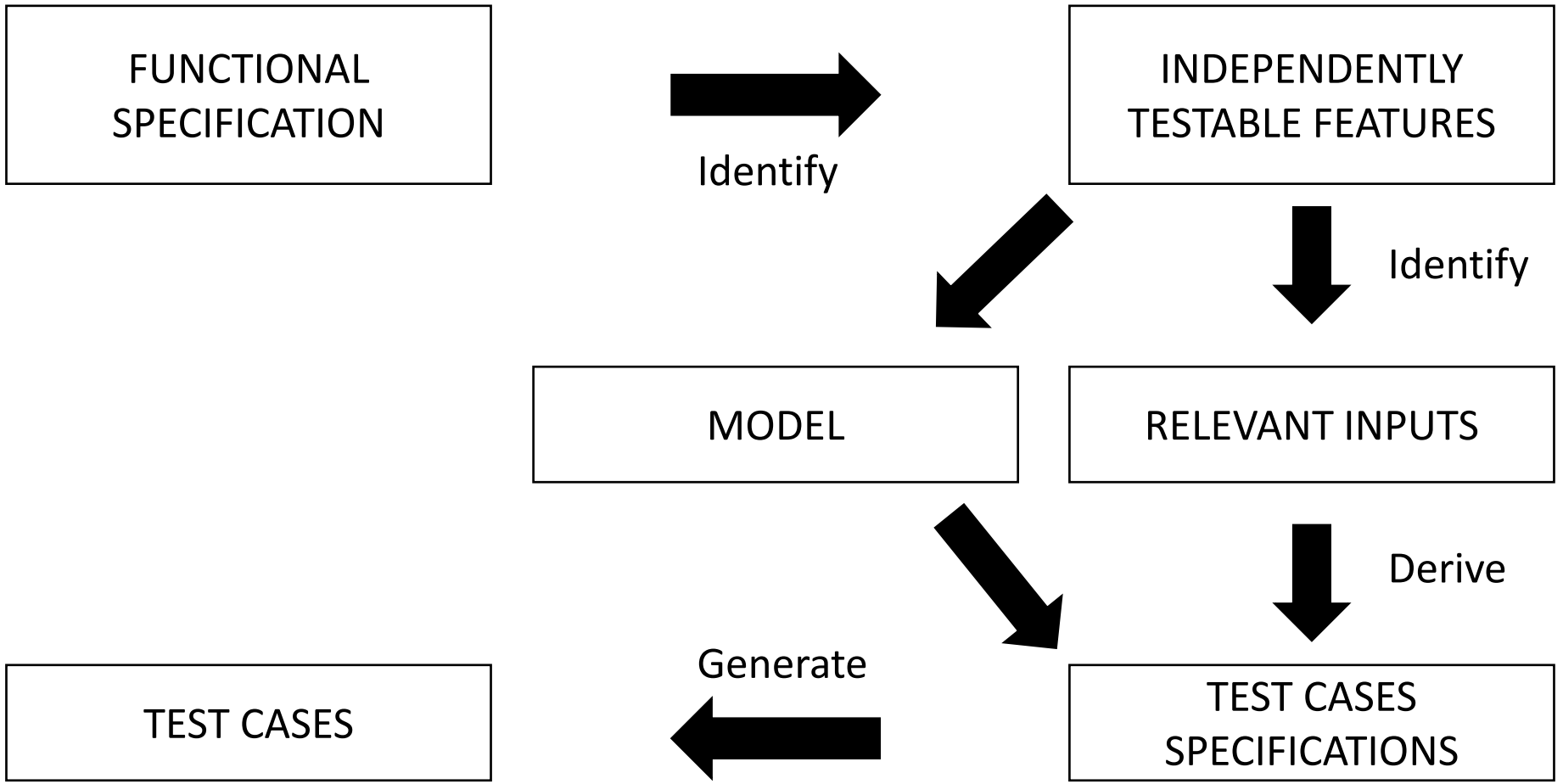
Abstraction is key – right balance between abstracting and expensive testing

Many other approaches
- Decision tables
- Flow graphs
- Historical Models
- …

Historical models – Distribution of security vulnerabilities in Firefox



Black-Box Testing



Decoupling; Automated Sub-tasks; Monitor testing process

Slide adapted from Alessandro Orso

Industry Standards Today

Automating black-box testing is essential for efficient, repeatable verification.

Tool / Approach	**Primary Use**
Selenium / Playwright	Automate browser interactions and end-to-end UI tests
Postman	Test REST APIs through HTTP requests, assertions, and collections
UFT / Ranorex	Automate functional and regression testing
AI-Assisted Testing	Generate test cases, scripts, and assertions; suggest edge cases

Techniques still matter: Category-Partition and Model-Based Testing help determine what to test; automation tools help execute those tests.

TOTT #7: AI-Assisted Black-Box API Testing

Using Postman and AI to Test a Spring Boot REST API

1. Apply specification-based testing to a working REST API.
2. Design and execute manual API tests using Postman.
3. Use AI to generate additional test assertions.
4. Evaluate test quality, correctness, and human review effort.

Problem Context: Online Checkout System

An online retailer needs a backend service to calculate the final amount customers pay. Inputs: Order subtotal, customer type, and promotional code. Business rules:

Rule	**Expected behavior**
Membership discount	Regular: 0%; Premium: 10%; Student: 5%
SAVE10	10% off when subtotal ≥ $75
FLAT20	$20 off when subtotal ≥ $100
Discount selection	Apply the larger eligible discount; no stacking
Discount cap	Maximum 25% of original subtotal
Shipping	$7.99 if discounted subtotal is below $50; otherwise free
Validation	Reject invalid inputs with HTTP 400
Rounding	Half-up to two decimal places

API Contract: Request and Expected Response

POST http://localhost:8081/api/discount

Request (JSON)

```
{
  "subtotal": 100.00,
  "customerType": "PREMIUM",
  "promoCode": "FLAT20"
}
```

Expected Response — HTTP 200

```
{
  "subtotal": 100.00,
  "discount": 20.00,
  "shipping": 0.00,
  "finalTotal": 80.00
}
```

Calculation: Premium $10 vs. FLAT20 $20 → apply $20 → free shipping → final total $80

Additional Boundary Tests

$74.99 + SAVE10	$0 discount; final $74.99
$75.00 + SAVE10	$7.50 discount; final $67.50
$55.55 / $55.56 + Premium	Shipping $7.99 / free

Impact of Prompting on AI-Generated Tests

Evaluation	Prompt 1: General Checks	Prompt 2: Specification-Aware
Generated output	5 assertions	17 new requests + tests
Strength	Reusable response checks	Systematic partitions, boundaries, invalid cases
Example	Arithmetic consistency	FLAT20 $99.99/$100; three customer types
Limitation	No business-rule context	Two incorrect shipping tests (12, 13), 2 redundant tests (9,11)
Human effort	Review assertions	Validate calculations and constraints

Takeaway: Postman executes our black-box API tests. Agent Mode can accelerate script writing, but neither a passing status code nor a well-formatted AI-generated assertion guarantees correct business behavior. Engineers should select the inputs and verify the test oracle.

Quizizz